Raspberry Pi Laparoscope

Mechatronic Project 478
Progress Report

Author: Pieter Willem Breytenbach
26122146

Supervisor: Prof K. Schreve

2026

Department of Mechanical and Mechatronic Engineering
Stellenbosch University
Private Bag X1, Matieland 7602, South Africa.

Plagiarism declaration

I have read and understand the Stellenbosch University Policy on Plagiarism and the definitions of plagiarism and self-plagiarism contained in the Policy [Plagiarism: The use of the ideas or material of others without acknowledgement, or the re-use of one's own previously evaluated or published material without acknowledgement or indication thereof (self-plagiarism or text-recycling)].

I also understand that direct translations are plagiarism, unless accompanied by an appropriate acknowledgement of the source. I also know that verbatim copy that has not been explicitly indicated as such, is plagiarism.

I know that plagiarism is a punishable offence and may be referred to the University's Central Disciplinary Committee (CDC) who has the authority to expel me for such an offence.

I know that plagiarism is harmful for the academic environment and that it has a negative impact on any profession.

Accordingly all quotations and contributions from any source whatsoever (including the internet) have been cited fully (acknowledged); further, all verbatim copies have been expressly indicated as such (e.g. through quotation marks) and the sources are cited fully.

I declare that, except where a source has been cited, the work contained in this assignment is my own work and that I have not previously (in its entirety or in part) submitted it for grading in this module/assignment or another module/assignment. I declare that have not allowed, and will not allow, anyone to use my work (in paper, graphics, electronic, verbal or any other format) with the intention of passing it off as his/her own work.

I know that a mark of zero may be awarded to assignments with plagiarism and also that no opportunity be given to submit an improved assignment.

Signature: 

Name: Pieter Breytenbach Student no: 26122146

Date: 18/02/2026

Artificial Intelligence (AI) Declaration

Declaration

- I have discussed the use (or non-use) of GenAI tools with my supervisor, and we have agreed on appropriate boundaries.
- I have verified and take responsibility for all information, arguments, data, and conclusions in this thesis/dissertation. I am accountable for the integrity of the work.
- I confirm that AI tools were not used in ways that breach SU's plagiarism, confidentiality, or research integrity policies.
- No data were generated by GenAI unless this was a deliberate and declared part of the research methodology.
- I am willing to be questioned in the oral examination on the scope, purpose, and verification of any GenAI use in this work.

Optional:

- I have included a reflective paragraph in my submission explaining the role (if any) of GenAI in my research process.

Institutional Acknowledgement

This declaration aligns with Stellenbosch University's Position Statement on the ethical use of AI in Research and Teaching-Learning-Assessment[1], the Draft interim SU guidelines on allowable AI use and academic integrity in assessment[2], and draws on international best practices.

[1] Position Statement on Ethical use of Artificial Intelligence in Research and Teaching-Learning Assessment

[2] Draft interim SU guidelines on allowable AI use and academic integrity in assessment

Executive summary

Title of Project
Raspberry Pi Laparoscope
Objectives
The objective of this project is to design, program, build and test a low-cost laparoscope.
What is current practice and what are its limitations?
Current laparoscope use in industry is expensive and not feasible to purchase for training and prototype testing of new laparoscopic medical robots.
What is new in this project?
For this project, instead of specialized lenses and camera modules, off-shelf products will be utilized to develop the product.
If the project is successful, how will it make a difference?
If this project is successful, it can save cost when designing and prototyping new medical robots. It can be utilized for training of new surgeons trainees.
What are the risks to the project being a success? Why is it expected to be successful?
The main risk of this project is delayed due to unavailability of hardware components. Therefore, the ordering of products is the main priority during early stages of the project.
What contributions have/will other students made/make?
Another student is performing the same project as me, thus notes and methods will be compared.
Which aspects of the project will carry on after completion and why?
The developed laparoscope will be utilized in future projects, where medical robots will be developed. The test station will also be utilized in testing future laparoscope designs.
What arrangements have been/will be made to expedite continuation?
To prevent delays, safety measures have been taken to prioritise the ordering of important hardware components.

Table of contents

List of figures

List of tables

Chapter 1

Introduction

1.1 Background

The Hippocratic oath is a fundamental principle that guides decisions and advancements in the medical profession. It motivates doctors and physicians to make ethical decisions to minimize patient harm. It also drives physicians to continuously improve practices to reduce surgical risks and complications. This principle has driven the discovery of minimally invasive surgery (MIS). (Ibrahim *et al.*, 2021)

Minimally invasive surgery has become the standard practice across surgical fields and is a rapidly growing division in the medical industry. One branch of this field is laparoscopy, which is used to perform advanced and complex surgical procedures. (Ibrahim *et al.*, 2021)

Laparoscopy is surgical approach where small incisions are made to allow insertion of specialized equipment into the patient's body. The primary instrument is the laparoscope, which is typically paired with other thin rod equipment based on the operations demands. Compared to traditional open surgery, laparoscopy offers advantages including faster recovery rates and reduced morbidity and patient discomfort, though it requires a steeper learning curve. These benefits translate to cost savings through shorter hospital stays and reduced analgesic requirements. (Subido *et al.*, 2018)

The laparoscope is a fundamental instrument to laparoscopic surgery. It consists of a thin rod with a camera attached to it at an angle, called the viewing angle, which is inserted into the patient's body. The captured footage is displayed in real-time to the surgical staff on a monitor. Based on the visible footage, the surgical staff make a diagnosis, perform procedures, and coordinate instrument positioning based on the visual feedback. (Ibrahim *et al.*, 2021) Figure 1.1 illustrates what the viewing angle is.

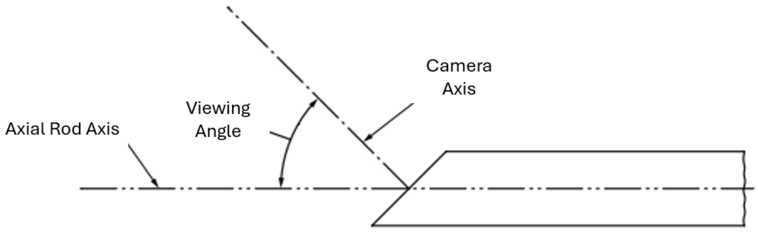


Figure 1.1: Illustration of the viewing angle

Recent advancements in laparoscopy focus on improving the laparoscope's visual capabilities through several approaches. These approaches include increasing the camera's field of view, implementing three-dimensional (3D) visualization, and incorporating artificial intelligence (AI) or machine vision technologies to captured footage. Additional advancement made to modern laparoscopes is the utilization of robotic manipulators, with 3 or more degrees of freedom (DOF) to assist the doctor during operations. (Subido *et al.*, 2018)

One of the main challenges of laparoscopic surgery is that it is a technically complex surgery to perform. Thus, inexperienced surgeons therefore require more training. However, laparoscopes are expensive equipment which makes them inaccessible for individual practitioners or small hospitals to purchase. (Xiao *et al.*, 2014)

Additionally, Stellenbosch University host an open or career day, where potential students can attend to see which possible degrees they can study at the university. Ideally, the university would appreciate having a working laparoscope, to demonstrate how laparoscopic surgeries are performed. Consequently, the laparoscope will be used by the attending students to experience how laparoscopic surgeries are performed on phantom bodies.

Therefore, there is a need for a low-cost laparoscope, which will be used for training and demonstration purposes and not intended for use on living patients.

1.2 Objectives

The aim of this project is to design, build and test a low-cost laparoscope.

This project's main objectives are to:

- Design, build and program a functional laparoscope by incorporating various Raspberry Pi components.
- Design a performance test to assess the laparoscope functionality and image quality abilities.
- Evaluate the developed laparoscope performance after performing the test.

1.3 Scope

To develop the laparoscope, there are various aspects that need to be considered. These aspects range from physical properties to software properties and all have different scopes sizes. However, the scope for this project is subject to change based on the timeline of the project or change of the desired objectives of the project.

An important scope boundary is that the developed laparoscope, should not meet all medical grade standards. Since, it will not be used on living patients, the product should not be made out of medical grade equipment or materials. However, it needs to function similarly to medical grade laparoscopes, henceforth, should meet certain functionality standards for laparoscopic equipment.

Additionally, the developed laparoscope does not require having any degrees of freedom (DOF). The viewing angle, angle between the optical axis of the camera and the laparoscope axial axis, should aim to be 30 degrees. It is not required to incorporate a DOF to the viewing angle of the laparoscope.

To determine the laparoscopes capabilities and properties a testing station is required. Since Stellenbosch University does not have a testing station for laparoscopic products, it is required to design and build one, as well as articulate the testing procedures for the laparoscope.

Considering the hardware components of the laparoscope, it is not within the project scope to manufacture hardware components from first principles. Rather, available hardware components will be purchased and repurposed if necessary to meet the project requirements.

Considering software of the laparoscope, it is required that the system should be able to only operate the different modules. Any additional software application, such as machine learning or machine vision, is not within the project's scope.

1.4 Motivation

As mentioned in Section 1.1, the main challenge with laparoscopic surgery is that it is a technically more complex procedure to perform, compared to open surgery. Thus, inexperienced surgeons require more training with the laparoscope and its additional equipment to achieve proficiency. However, laparoscopes are expensive surgical equipment, that isn't affordable investment for individuals or small medical facilities.

This project could also add value to the development of new medical robots. Most preliminary designs of medical robots require laparoscopes that can evaluate the robot's capabilities and flaws. However, they do not require expensive medically approved laparoscopes, rather, low-cost alternatives are preferable, as they minimize financial loss if damaged during testing or development.

Additionally, a low-cost laparoscope can be used during future career days to demonstrate how laparoscopic surgeries can be performed for prospective students.

1.5 Use of AI Tools

Anthropic is the main Artificial Intelligence (AI) company that will be used for this project, specifically its Claude Sonnet 4.5 model. This AI model assisted with research papers identification, sentence construction, grammar checks and code debugging. All AI-generated material will be reviewed and verified by the student to ensure the integrity and originality of the provided work.

1.6 Proposal Overview

This report aims to enlighten the reader of the various sections of work performed to complete this project. Chapter *2* informs of the history of laparoscopy, afterwards it explains various segments of the work required to understand completing the project. Chapter 3 aims to provide an overview of the design process for the laparoscope, while chapter 4 explains how a digital model was created of the laparoscope, to simulate how heat is transferred within the product. The following chapter 5 highlights all important test that were executed for the laparoscope. The last chapter of this report, chapter *6*, provides a final motivation why this project was performed.

Additionally, in the appendixes, information is given regarding the scheduling of the project, the GA self-assessment, the planned work hours, and assembly drawings of the laparoscope.

Chapter 2

Literature Review

A literature review has been conducted for this project. This section of the report documents the relevant research performed and divides into to the following sections:

- **Laparoscopy Overview:** An overview of what laparoscopy is and its history.
- **Current Technological Advancements:** The current directions that modern laparoscopy is improving on.
- **Current Available Technology:** A summary of current Laparoscopes that is available in South Africa.
- **Available Hardware Components:** An overview of different hardware modules available for this project.
- **Software Considerations:** An brief overview of the available programs that can be used for this project.
- **Visual Inspection Considerations:** An overview of visual inspection and aspects of camera that need to be mastered.

2.1 Laparoscopy Overview

Laparoscopy is a branch of endoscopy. It has become an integral part of surgical procedures in all surgical fields in recent decades. Inspired by the minimally invasive surgery (MIS) approach, laparoscopic surgery consists of making one or more small incisions in the abdomen of the patient. Surgeons use these incisions to insert thin, rod medical equipment into the patient to perform the surgery. (Ibrahim *et al.*, 2021)

An essential tool of this surgical procedure is the laparoscope. Due to the small incisions, the surgeons' view is limited. The laparoscope is the solution to this problem. It allows surgeons to view the interior within the patient as the operation is being performed. Therefore, they can clearly manipulate their equipment from outside the patient to perform complex procedures inside the patient. (Subido *et al.*, 2018)

Modern laparoscopes are equipped with small cameras and light sources at the end of the tool. This camera is inserted into the patient, and the captured footage is displayed immediately to the surgical staff on a monitor. With the visible footage, the surgical staff can make a diagnosis and perform operations. (Subido *et al.*, 2018)

However, the first laparoscope was invented almost a century ago. They did not consist of cameras, but rather, of complex lens assemblies at the end of the laparoscope and cold light sources. They were used in a similar method to a microscope. However, this severely limited the surgeon's ability to operate and the surgical staff's ability to assist the operating surgeon. (Ibrahim *et al.*, 2021) Figure 2.1 illustrates how the original laparoscope was designed.

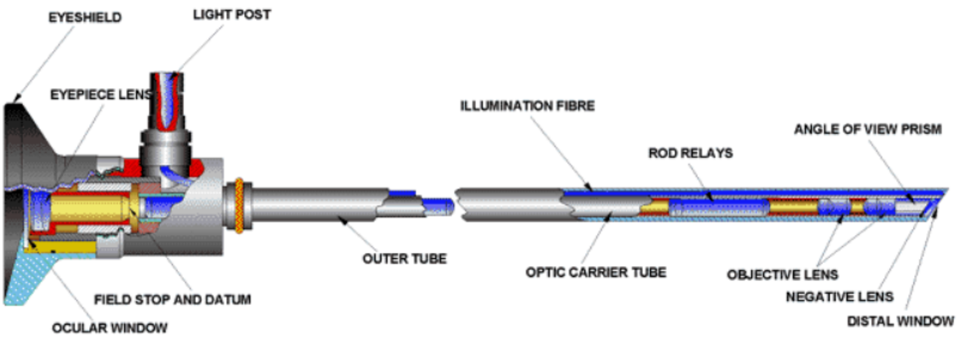


Figure 2.1: Original laparoscope design. (EBME and Clinical Engineering Articles, 2026)

Laparoscopic surgery has been one of the greatest advancements in the medical industry in recent years. Buia *et al.* (2015) shows that laparoscopic surgery has a quicker recovery time and less pain after surgery compared to traditional open surgery. However, it shows that certain procedures are more time-consuming and complex to perform.

2.2 Current Technological Advancements

Since the addition of cameras to laparoscopes, there have been significant opportunities for advancements. These advancements range from hardware improvements to software additions to the existing design.

The main hardware advancement for laparoscopes is the improvement of camera modules. However, more advancements have been made with the addition of robotics to the laparoscopes, which have greatly assisted the laparoscopic surgical industry. Robotic manipulators are utilized to assist surgeons by maintaining stable control of the laparoscope. Therefore, ensuring consistent and clear visualization during the procedure. (Subido *et al.*, 2018)

There are multiple additions currently being made to the software application of laparoscopes. Subido *et al.* (2018) list the following additions:

- Digital imaging enhancement techniques.
- AI incorporation.
- 3D visualization.
- Object tracking.

2.3 Current Available Technology

There is a range of laparoscope or endoscope suppliers across the globe. Each of these suppliers produces more than one product. Therefore, only local suppliers will be reviewed, as well as Intuitive Surgical, which produces the *Da Vinci* system, which has dominated the surgical robot market. (Subido *et al.*, 2018)

One of the main suppliers of laparoscopic equipment in South Africa is Karl Storz Endoskope, a German company, with a sales office in Cape Town. They produce numerous products in the medical industry. (Karl Storz Endoskope, 2026*a*) One of their products is the *TIPCAM1 RUBINA*, illustrated in Figure 2.2 below.

TIPCAM1 RUBINA is a rigid endoscope with a viewing angle of either 0° or 30°. It comes with automatic horizon control and the ability to easily swap between 2D and 3D imaging. It has an outer diameter of 10.3 mm and a working length of 320 mm. Additionally, it has 4K image resolution and a mass of 420 g. (Karl Storz Endoskope, 2026*b*)

Figure 2.2: Figure of the TIPCAM1 RUBINA. (Karl Storz Endoskope, 2026*b*)

Olympus Corporation is another company that supplies medical equipment globally. They are a Japanese manufacturing company that produces the *ENDOEYE 3D 10 mm*, which can be seen in Figure 2.3 below. The *ENDOEYE 3D 10 mm* is a video telescope, with the ability of 3D imaging at a viewing angle of 0° and

Figure 2.3: Figure of the ENDOEYE 3D 10 mm. (Olympus Corporation, 2026)

30°. It is able to perform image rotation at 30° viewing angle, while maintaining the horizon. (Olympus Corporation, 2026)

Certain international companies use local distributors to sell their products. Intuitive Surgery, an American company, uses this method. They designed and manufacture the *Da Vinci* medical robot. This system has various variants for different surgical purposes. Currently, there is *Da Vinci Xi* and *Da Vinci X* for multi-port surgery. However, the *Da Vinci SP* is designed for single port surgery. These robot systems are equipped with seven DOF movement and 3D visualization. Their main drawback is their large size and enormous financial procurement and maintenance cost. (Intuitive Surgical, 2026) Below in Figure 2.4 of the *Da Vinci SP*.

Figure 2.4: Figure of the Da Vinci SP. (Intuitive Surgical, 2026)

2.4 Available Hardware Components

There are various hardware components required to build a laparoscope. Therefore, this section will be divided into multiple subsection where different products will be reviewed for each component. These subsections are:

- Single board computer module.
- Camera module.
- Lighting module.

2.4.1 Single Board Computer Module

Single board computers (SBC) are a credit-card-size computer, able to perform complex computations while operating various systems. For the laparoscope, it will be the main "brain" of the system. Therefore, it is critical that it can perform all necessary tasks it requires. There are two main suppliers of SBC:

- Raspberry Pi Ltd.
- Arduino AG.

Raspberry Pi has a large range of products that is available for the project. However, Raspberry Pi's SBCs has an important feature installed into most of SBC, which would be beneficial to the project. This feature is the designated camera serial interface (CSI) or the MIPI camera display transceivers, according to Raspberry Pi. This feature allows for high-quality videos at high frame rates. (Raspberry Pi Ltd, 2026*a*) Table 2.1 below shows the available options of all the Raspberry Pi SBC:

Table 2.1: Raspberry Pi SBCs. (Raspberry Pi Ltd, 2026*b*)

Raspberry Pi SBC	RAM	CPU Speed	MIPI Ports	Cost
Pi 5	1-16 GB	2.4 GHz	2	R 3 266.00
Pi 4 Model B	1-8 GB	1.5 GHz	1	R 1 360.00
Pi 3 Model B+	1 GB	1.4 GHz	1	R 640.43
Pi Zero 2 W	512 MB	1 GHz	2	R 240.16
Pi Zero 2	512 MB	1 GHz	2	R 309.90

All the SBC listed in Table 2.1 above have these features:

- 40 General-Purpose Input/Output (GPIO) Pins.
- Wi-Fi Capabilities.
- Bluetooth Capabilities.

- USB Ports.

Arduino, similar to Raspberry Pi, have a wide range of products available for this project. However, they lack a CSI port. Therefore, they are incompatible with the Raspberry Pi Camera Modules. However, they can use other camera modules, with the use of SPI and I2C GPIO pins. Although this options isn't favourable, due to trade-off in quality during longer videos, it is still a feasible option that can be explored. (ArduCAM, 2022) The Table 2.2 below shows the available products from Arduino:

Table 2.2: Arduino SBCs. (Arduino AG, 2026*b*)

Arduino SBC	RAM	CPU Speed	SPI Ports	Cost
UNO R4 WiFi	32 kB	48 MHz	1	R 488.46
UNO R4 Minima	32 kB	48 MHz	1	R 352.33
UNO R3	2 kB	16 MHz	1	R 469.24
GIGA R1 WiFi	1 MB	480 MHz	2	R 1 247.58
Nano ESP32	512 kB	240 MHz	1	R 358.74
Nano RP2040 Connect	264 kB	133 MHz	4	R 389.17
UNO Q	2 GB	2 GHz	1	R 1 035.54
Nano 33 BLE	256 kB	64 MHz	4	R 456.43
Nano 33 IoT	520 kB	240 MHz	6	R 427.6
Nano Every	6 kB	20 MHz	1	R 232.22

2.4.2 Camera Module

The camera module is an important component for this project that requires careful considerations when selecting. It is the main component that will be inserted into a patient's body to capture the real-time images for the surgical staff to perform operations. Therefore, the quality of the camera module will dictate the performance of the entire laparoscope.

The camera module has the most stringent physical specifications of all the components. The entire module must fit within a Ø10 mm hole. This proved to be the largest failing criterion for camera modules available on the market, due to large camera assemblies; or the large printed circuit board (PCB) attached to the camera assemblies. Figure 2.5 below demonstrates how the PCB attached to the camera module is to large fit inside the Ø10 mm hole. (ArduCam, 2020)

The second most common failing criteria for camera modules were the focus depth of the camera modules. Since the distance from the camera module lens and focus object is small, 5 mm to 100 mm, camera modules with large focus depth ranges are not practical for this project. Therefore, eliminating products from the hummingbird mobile camera module series, developed by Leopard

Figure 2.5: Figure of Raspberry Pi camera module 3. (Raspberry Pi Ltd, 2026*a*)

Imaging, (Leopard Imaging, 2026) and the OV5647 spy camera module, developed by ArduCam. (Arduino AG, 2026*b*)

Based on these two main selection criteria, two products have been selected to use for this project:

- DF Robot 3-in-1 endoscope camera, produced by DF Robot.
- 1 MP 3.5 mm OV9734 HD endoscope medical camera module, produced by Vision Meta.

Table 2.3 compares the two product mentioned above, while Figure 2.6a and Figure 2.6b illustrate the products.

Table 2.3: DF Robot compared to Vision Meta.

Feature	**DF Robot**	**Vision Meta**
Diameter (mm)	8	3.5
Field of View (°)	70	90
Depth Focus (mm)	N/A	5–100
Resolution	0.864 MP	0.922 MP
Built-in LED	Yes	No
Cost	R 457.70	R 995.96 ($60.00)
Shipping Cost	R0.00	R 1 344.55 ($81.00)

(a) DF Robot endoscope module. (Micro Robotics (Pty) Ltd, 2026)

(b) Vision Meta endoscope module. (Shenzhen VisionMeta Technology Co., Ltd, 2026)

Figure 2.6: Illustration of camera modules

These camera modules have USB compatibility and meets the physical specifications for this project and would in theory work. However, a feature that laparoscopes have is that the viewing angle is 30°. Due to the DF Robot endoscope's large diameter and length, this feature makes it unpractical for this project, unless the module is modified.

2.4.3 Lighting Module

Due to the laparoscopes being inserted into a patient's body, there will be limited illumination within the patient, which will impact image quality. This can be easily resolved by installing illumination hardware to the laparoscope. These hardware components should be compact, while producing enough illumination, which is measured in luminous intensity (cmd). Micro Robotics, has two main products that can be used for this project. Both products are listed in Table 2.4 and illustrated in Figure 2.7 below. (Micro Robotics (Pty) Ltd, 2026)

Table 2.4: List of illumination hardware. (Micro Robotics (Pty) Ltd, 2026)

Product	Luminous Intensity	Dimensions
LED SMD 1206	200-850 cmd	3.2 x 1.6 x 1.1 mm
LED SMD 0603	45-180 cmd	1.6 x 0.8 x 0.55 mm

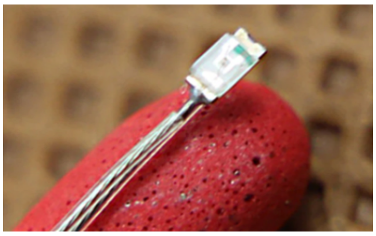


Figure 2.7: Figure of LED SMD 0603. (Micro Robotics (Pty) Ltd, 2026)

2.5 Software Considerations

The most important software consideration of this project is that of the camera module interface. Since, both camera modules that are going to be used in this project, use USB interface, the recommended software to operate the camera modules is AMCap. (Micro Robotics (Pty) Ltd, 2026) However, Open CV can also be utilized to control USB camera modules.

The software language is an important aspect that needs to be considered. However, this is dependent on the SBC used. If a Raspberry Pi is used, then Python is recommended language to use. (Raspberry Pi Ltd, 2026*a*) However, if an Arduino is used then the recommended language to use is either C or C++. (Arduino AG, 2026*a*) Visual Studio Code (VS Code) can be used to perform all programming for this project.

2.6 Visual Inspection Considerations

This project requires a camera to be a functional laparoscope. However, there are various factors to be considered to take high-quality images. This section of the literature reviews aims to explain each of these factors. Consequently, how they can impact this project.

Important concept to understand, is how a digital image is captured. A camera sensor consists of a matrix of millions of light cavities. When a picture is being taken, these cavities, called "photosites", are exposed to the photons of the object or scene. The photons are then collected by the photosites and stored as an electrical charge. Consequently, when the image is finished being taken, these charges are quantified and then processed to form the image. (Cambridge in Colour, 2020*a*)

One aspect that needs to be mastered to captured quality images is the exposure triangle. It is 3 pillars that dictates the quality of an image. The first pillar, called aperture, determines the size of the area opened in the lens which light can filter through to sensor. The larger the aperture value, measured in f-stop values, the smaller the hole size where light filters through the lens. These pillars relate to the depth of field of the images. The smaller aperture correlates to shallower depth of field images. (Cambridge in Colour, 2020*b*)

Shutter speed is the second pillar of the exposure triangle. When a digital images is being taken the lens opens up to expose the photosites to the light. Consequently, shutter speed refers to total duration of the lens being open. If the shutter speed increases, then the sensors is exposed to reduced duration of light from the images. Therefore, as the shutter speed increases, the images will gain more quality, with more definitive objects that is more ”frozen” in the image. (Cambridge in Colour, 2020*b*)

The third pillar is ISO. This determines how sensitive the sensor is to the exposed light. As the ISO value increases, the higher the sensitivity value will become. Therefore, the image will have increased levels of noise captured if the ISO value is increased. (Cambridge in Colour, 2020*b*)

Focal length is a critical feature that has to be considered for this project. Focal length, is the distance between the camera sensor and the camera lens. As this value increases, then the field of view value will decrease. Field of view, measured in degrees, is essentially how much of an image can be seen. If this value increases, then a wider image can be taken. (Morrow, 2026)

Chapter 3

Design

This chapter of the reports documents the design process for the laparoscope. From determining the design requirement, to concept designs generation, and detail design of the laparoscope.

3.1 Design Requirements for Laparoscope

As mentioned above, this chapter documents the list of requirements that needs to be determined to be able to design a functional laparoscope. These requirements are all listed in the tables below, as well as depicted in the diagrams below.

3.1.1 Stakeholder Analysis

The stakeholder analysis is shown in Table 3.1 below:

Table 3.1: Stakeholder analysis

Stakeholder ID	Stakeholder	Classification	Description	Life Cycle Stages	Key Interest/Needs
SH_01	Dr Diayar	Indirect	The doctor that recommend the development of the system	Concept	The functional development of the system

Continued on next page...

Stakeholder ID	Stakeholder	Classification	Description	Life Cycle Stages	Key Interest/Needs
SH_02	Prof Schreve	Secondary	Supervisor of the project	All stages	The functional development of the system, and the engineering design methodology followed to develop the system
SH_03	Medical trainee	Primary	The personal that will use the developed system to improve their surgical abilities	Operation	The low-cost production of the system and the easy operations of the system
SH_04	Hardware suppliers	Secondary	The suppliers of the hardware components for the system	Design, Installation	Technical specifications of the hardware components
SH_05	Structural manufacturers	Secondary	Personnel responsible for the construction of structural components of the system	Design, Installation	Technical specifications of the structural components
SH_06	Stellenbosch University	Secondary	Main financial supplier and main management team of the project	All stages	The cost of the project, academic procedures followed and safe operations during all stages of the project

Continued on next page...

Stakeholder ID	Stakeholder	Classification	Description	Life Cycle Stages	Key Interest/Needs
SH_07	Future researchers	Primary	Future personal that will use the developed system or select components of the system to perform further researchers	Maintenance, Operation	The technical conditions of the developed system, the technical conditions of systems hardware components
SH_08	Construction and maintenance staff	Secondary	Personnel responsible for the construction and maintenance for the entire system	Installation, Maintenance	System design and building procedures, systems maintenance procedures
SH_09	Design team	Secondary	Personnel responsible for the design of the entire system	Design	System requirement and hardware physical properties
SH_10	Software team	Secondary	Personnel for the software component of the entire system	Design, Installation	System requirements and hardware software capabilities
SH_11	Potential student attendees	Primary	Potential students that will use the system to perform surgical procedures	Operation	Easy operations and usability of the system

3.1.2 Design Independent Requirements

The design independent requirements (DIP) requirements for the laparoscope can be seen in Table 3.2 below. Additionally, how the DIPs interface from the narrower system of interest (NSOI) with the wider system of interest (WSOI) and environment can be seen in Figure 3.1 below:

Table 3.2: Design independent parameters

DIP_ID	Design Independent Parameter Description	Value / Range	Reference
DIP_01	Laparoscope maximum diameter	Ø < 10 mm	Supervisor
DIP_02	Grid electricity availability	230 V, 15 Ampere at 50 Hz	World Standards (2025)
DIP_03	Ambient operating temperatures range (Stellenbosch)	2 C° - 42 C°	Weather & Climate (2026)
DIP_04	Laparoscopes total building cost	R 6000.00	Stellenbosch University (2026)
DIP_05 [1]	Laparoscope illuminance produced	40 000 LUX - 160 000 LUX	International Electrotechnical Commission (2021)
DIP_06	Relative maximum humidity in operating rooms	< 81%	Weather & Climate (2026)
DIP_07	Maximum deviation of specified viewing angle	10°	International Organization for Standardization (2025)
DIP_08	Frequency of real-time digital images updates	30 Hz - 60 Hz	Karl Storz Endoskope (2026*b*)

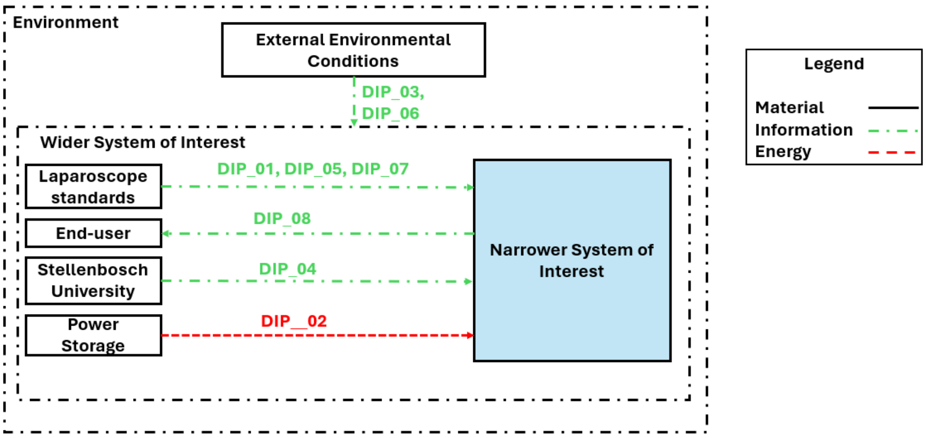


Figure 3.1: System boundary

[1]There is not a standard for the required illuminance to be produced by a laparoscope or an endoscope. However, operating rooms require an illuminance value between 40 000 and 160 000 LUX, which is the value that will be designed for.

3.1.3 Functional and Non-Functional Requirements

Below in Table 3.3 is the functional requirements for the laparoscope. Consequently, the Table 3.4 contains the non-functional requirements.

Table 3.3: Functional requirements

FR_ID	Functional Requirement Description	Primary/Derived
FR_1	User interface	Primary
FR_2	System processor	Primary
FR_3	Power supply	Primary
FR_4	Operate Camera Module	Primary
FR_4.1	Specify camera settings	Derived
FR_4.2	Specify light intensity	Derived
FR_4.3	Capture digital image	Derived
FR_5.0	Display real-time images	Primary
FR_6.0	Store system data	Primary

Table 3.4: Non-functional requirements

NFR_ID	Non-Functional Requirement	Description
NFR_01	Relevant compliance	The design of the system should meet the relevant prescribed standard for this project, to ensure that the laparoscope meets all physical capabilities and functionality of a real laparoscope
NFR_02	Documentation	The design, implementation, construction, maintenance and testing procedures for the system should be thoroughly documented
NFR_03	Availability	The components for this project should be available to purchase if a component breaks or should be replaced
NFR_04	Adjustable	The system design should be developed that distinct features can be changed without creating large external costs
NFR_05	Usability	System should be easy to operate for limited-training users

3.1.4 Technical Performance Measures

The technical performance measures (TPM) can be seen in the Table 3.5 below, based on the requirements in Table 3.3. The reasoning behind all the TPM target values can be found below the table.

Table 3.5: Technical performance measures

FR_ID	Function	TPM_ID	TPM Description	TPM Target Value(s)
FR_1.0	User interface	TPM_01	Minimum user interface input(s)	≥ 1 input
		TPM_02	Minimum user interface signal(s)	≥ 1 signal light
FR_2.0	System processor	TPM_03	Minimum GPIO pins	≥ 1 GPIO Pin
FR_3.0	Power supply	TPM_04	System start up cycle duration	≤ 2 s
		TPM_05	Total duration of power supply required	≥ 6.7 hours
FR_4.0	Operate camera module	TPM_06	Amount of camera modules	1 camera modules
		TPM_07	Working length of the system	300 mm
		TPM_08	Viewing angle of camera module	30°
FR_4.1	Configure camera properties	TPM_09	Focus depth of the camera module	30 mm - 80 mm
FR_4.2	Configure light intensity	TPM_10	Illumination of system	40 000 LUX - 160 000 LUX
		TPM_11	Maximum surface temperature	≤ 41 °C
FR_4.3	Capture digital image	TPM_12	Minimum field of view	≥ 70°

Continued on next page...

FR_ID	Function	TPM_ID	TPM Description	TPM Target Value(s)
		TPM_13	Minimum resolution	460 x 460 pixels
FR_5.0	Display real-time images	TPM_14	Minimum resolution of display	460 x 460 pixels
		TPM_15	Maximum latency	0.5 s
FR_6.0	Store system data	TPM_16	Storage capacity	68.56 GB

- **TPM_01:** The bare minimum for the laparoscope to function properly with user inputs is to at least install an on or off switch. It is the bare minimum required for this project.

- **TPM_02:** For the operator to know the laparoscope is still functioning properly, without looking at the display screen or light module at the end of the laparoscope, there should be at least 1 signal light on the laparoscope to indicate that the system is receiving power.

- **TPM_03:** To operate the signal lights the microcontroller or SBC will require GPIO pin to push current into the signal lights. Thus, the minimum required is the minimum of the laparoscope signal lights.

- **TPM_04:** This is dependent on the component selected and the different interfaces selected between the different components. However, the desired start-up cycle duration should be less than two seconds.

- **TPM_05:** The system should be able to power on for long durations of times. The total amount of time it should be powered on is the duration of a typical laparoscopic surgery.

- **TPM_06:** The system requires at least one laparoscopic camera. A second camera module can help the system achieve 3D visualization. Any additional cameras would be redundant for this project.

- **TPM_07:** The system design should be similar to current laparoscopes. Current laparoscope designs has a working distance of 300 mm. The working distance is the distance of the rod that can be inserted into the patient's body.

- **TPM_08:** Different laparoscope products have different viewing angles. Some has a viewing angle of 0°, while there are products that have a viewing angle of 90°. However, a common viewing angle is 30°, which is the value that will be pursued in this project.

- **TPM_09:** The distance between the camera sensor of the laparoscope and the object that is being captured is the focus depth.
- **TPM_10:** The total illumination required per medical theatres.
- **TPM_11:** The maximum temperature of the casing of the laparoscope tip, based on medical standards.
- **TPM_12:** Smallest viewing angle applied to a laparoscopic commercial product is 70°, while certain laparoscopes has a viewing angle larger than 70°.
- **TPM_13:** The minimum resolution for the camera module is determined by the first determining the maximum visible seen at the smallest fov, and multiply that value with how many pixels a millimetre area should represent:

$$Pixel = Focusdepth_(max) * sin(fov/2) * 2 * 5pixels/mm \quad (3.1)$$

- **TPM_14:** The minimum resolution of the display is the same as the minimum resolution of the camera.
- **TPM_15:** This value is subject to change based on the different components used for the system. Therefore, it is important to minimize the latency of each component or interaction between different components. 0.5 s is just an absolute maximum latency that will be accepted.
- **TPM_16:** If the minimum resolution an image is 460 x 460 with RGB colour format, then 1 image requires 634.8 KB. (PxP*24/8) If the frame rate is 30 fps and this continued for 60 minutes, then a total space of 68.5584 Gb is required.

3.2 Concept Generation

For this project three concepts were generated, based on the principle that the design utilizes the DF Robot endoscope, which has the dimensions of Ø8 mm and a total length of 25 mm. This module was chosen, since it is the maximum space required to fit all the necessary PCB and camera assembly to create an endoscope, that still fit within the Ø10 mm hole. Each concept is designed to have a viewing angle of 30°.

The first concept, relies on bending the shaft that the endoscope camera is attached to. However, the total diameter of the laparoscope within the skin and muscle tissue of the patient should remain under the Ø10 mm limit since the thickness of the muscle and tissue of patients are not all the same, the design has to be made for the worst case scenario, which is for an obese patient around 10 cm. Figure 3.2 below illustrates how the design would look like:

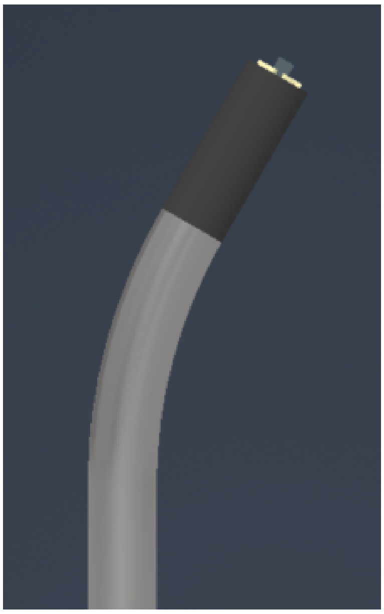


Figure 3.2: Concept 1: Bent shaft design

The second concept, rely on that the lens assembly is tilted to 30° relative to the axial axis of the laparoscope. This concept, does require careful hardware consideration and positioning, however is a proven method in the industry. Figure 3.3 illustrates this design:

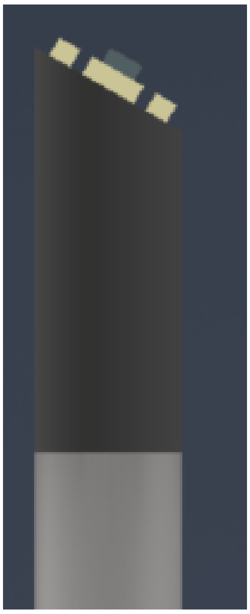


Figure 3.3: Concept 2: Tilted lens design

The last concept, relies on a pulley system, which allows the camera to be tilted at a desired angle by the practitioner, while the operation is being performed. This method relies on fine detail design, but can prove to be beneficial for future research purposes. The final concept is illustrated in Figure 3.4 below:

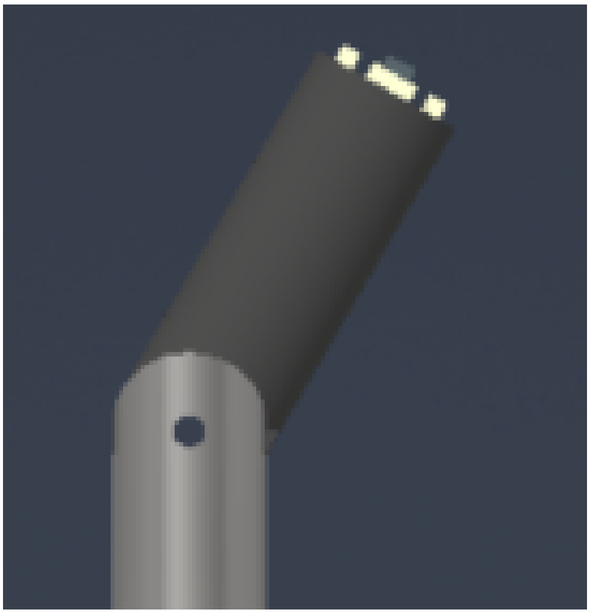


Figure 3.4: Concept 3: Adjustable viewing angle design

All three concepts mentioned above can be used for the project, since they meet all physical demands of the laparoscope. However, each concept has its defects. Concept one and three both have a defect when their camera is tilted at 30°. When the laparoscope is being rotated, there might be vital information that is missed in the direction of the laparoscope's axial axis, due to the camera lenses not being centred on the axis itself. It can also possibly cause damage to the patients, if the laparoscopes are being turned and accidentally pushed into the patient's internal organs.

Due to the reason stated above, the second concept is the preferred concept that will be further developed.

3.3 Detailed Drawing

Based on the previous section, it was determined to use the tilted camera design. A common problem with laparoscope and endoscope design is the small scale of the design. There is limited commercial products for such small scale design. Consequently, it leads to the design and manufacturing of custom parts that can perform the required duties, with size constraints. This design, however, adds additional complexity with the camera module and light source of the laparoscope being mounted to the tip of the laparoscope. These two components generate heat, which should not be transferred into the "patient's" body. Therefore, the design should account for this by minimizing the total heat transferred into the ”patient’s” body or by maximizing the total heat transferred through the laparoscope outside the ”patient’s” body.

This design does also reduce the complexity of the design, due to placements of the camera module and light source. For conventional laparoscope, either the camera module or light source or both, are placed outside the patients body. This design requires then complex lens assemblies for the camera module to be able to view inside the ”patient’s” body with the addition of plastic optical fiber cables for the light source.

To design for the heat problem, there are two main methods to control the heat dissipation of the camera module and light source:

- Actively cool down the heat sources while they are being operated.
- Actively conduct the heat from the heat source, located at the tip, to the base of the laparoscope, where the heat can be easier dissipated.

The first method, however functional, is not a practical solution for this design, due to the size constraints. For this method to work, cold air should be blown to the tip of the laparoscope, while warm air should also be extracted from the tip. Therefore, 2 ventilation tubes should lead down from the handle of the laparoscope to the tip of the laparoscope, while being confined in a Ø 8 mm hole. Additionally, it should provide space for the wiring of the camera module and heat sources. It is important that air should both be extracted and provided to the tip of the laparoscope, from the handle, and cannot be extracted from or dispersed into the ”patient’s” body. Otherwise, this design will not function as a medical laparoscope. Due to the limitations stated above, this design is not feasible.

The second method, however, has been accomplished. A patent has been filed with the United States Patent and Trademark Office, (USPTO) which utilized heat pipes to conduct the heat from the light source at the tip of the laparoscope to a heat sink in the handle of the laparoscope. (Schrader *et al.*, 2016) However, this patent is only designed to conduct heat generate by the light source, while the design that will be pursued will also account for the laparoscope camera module heat generation.

A critical feature of this design is the physical hardware. Heat pipes and heat sinks are required to be sufficiently conduct the heat as well as properly dissipate the conducted heat safely and efficiently out of the product. Fortunately, RS Components Limited (RS) provides both heat pipes and heat sinks in a wide variety of sizes. Importantly, the smallest diameter heat pipes they sell are 3 mm, at a maximum length of 250 mm. However, TPM_07 dictates that the working length of the laparoscope should be 300 mm. Therefore, a combination of heat pipes will be required to conduct the generated heat from the tip all the way to the heat sinks, through means of a coupler.

Another coupler is required to attach at the tip of furthest heat pipe from the handle, to be able to conduct the generated heat from the camera module and light source properly to the heat pipes. However, this coupler will require additional modification to account for the PCB of the camera module and light source. It requires a design balance to assure that there is enough material to properly conduct the heat to the heat pipes, while allowing the camera module and light source to operate properly and is not constricted by the design.

A Stainless steel pipe will be utilized to acts as a cover for the entire laparoscope's working length, keeping everything together and protected. Stainless steel was selected due to it being the only commercial material that is produced with such small dimensions.

For the handle of the laparoscope, 3D printing was selected to create a custom handle for the laparoscope, due to its ability to create custom components. However, an important decision was made to use acrylonitrile butadiene styrene (ABS) filament, instead of polylactic acid (PLA) filament. ABS filament, although harder to print, has higher temperature tolerances and can withstand mechanical loads better.

Figure 3.5 below shows the full assembly of the laparoscope, while Figure 3.6, shows the assembly of the main body of the laparoscope. The full assemblies of these design can be seen in the Appendix 6.1, with the drawing tree as well.

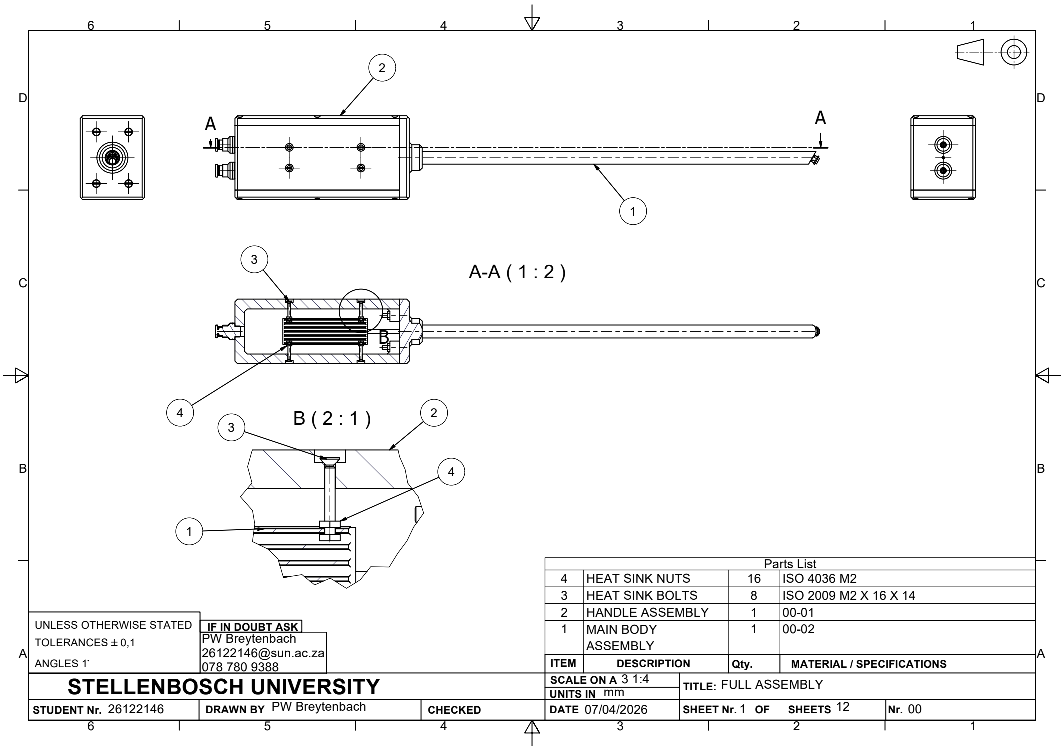


Figure 3.5: Full assembly drawing

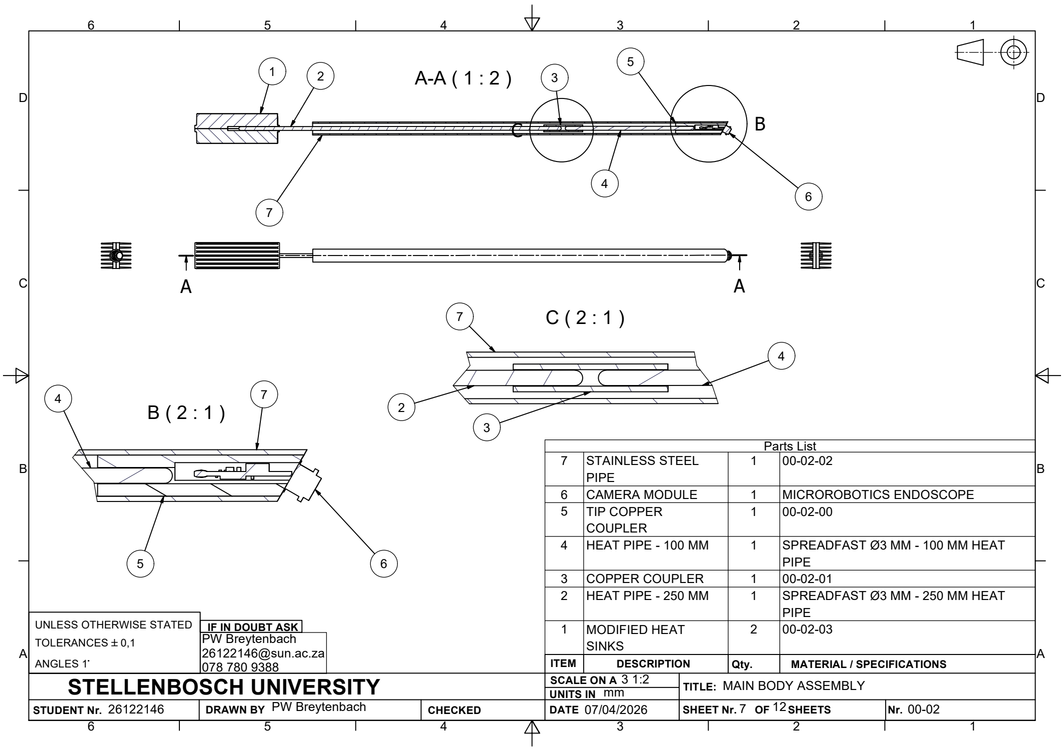


Figure 3.6: Main body assembly drawing

Chapter 4

Heat Transfer Evaluation

Chapter 5

Laparoscope Testing

In this chapter of the report, all the documentation of the test performed for this skripsie can be found. The tests performed can be divided into three section: validation test; specification test; and functionality test.

5.1 Validation Tests

The validation tests are tests to determine whether a calculated theoretical value is accurate or not. Currently, there are four validation tests that need to be performed. These tests consist of:

- Field of view validation test
- Viewing angle validation test
- Produced illuminance validation test
- Heat transfer model validation test

The first two test are test created by the International Organization for Standardization, to determine the field of view and viewing angle of the laparoscope. ISO 8600-3 describes the test procedures. Additionally, the standard gives the allowable deviation from the theoretical values. For the field of view, the value cannot deviate more than 15 % and for the viewing angle it cannot deviate more than 10°. Table 5.1 below shows the results of these test:

Table 5.1: ISO standard test results

Test	Theoretical Value	Actual Value	Deviation
Field of view	90°	T.B.D.	T.B.D
Viewing angle	30°	T.B.D.	T.B.D

The third test is to determine how much illuminance is produced by the light source and whether the value is within the expected range. This value is within a range, due to the nature of the selected components for light source. Four White SMD 1206 LEDs were selected to create the light source and according to their datasheet, they have a luminous intensity value between 200 mcd and 850 mcd and a viewing angle of 120°. The equations below shows how the theoretical illuminance can be calculated: (Zumtobel, 2018)

$$\Omega = 2 * pi * (1 - cos(\Theta/2)) \tag{5.1a}$$

$$\Phi = I * \Omega \tag{5.1b}$$

$$r = d * tan(\Theta/2) \tag{5.1c}$$

$$A = pi * r^2 \tag{5.1d}$$

$$E = \Phi/A \tag{5.1e}$$

Here are the list of all the variable used in the equations above:

- Ω: Beam solid angle (sr)
- Θ: Viewing angle of LED (°)
- Φ: Luminous flux (lm)
- r: Radius of the illuminated area (m)
- d: Distance between the light source and the illuminated area (m)
- A: Area of the illuminated area (m^2)
- E: Illuminance (LUX)

Based on the equation above, Table 5.2 below are the expected ranges for the developed product and the measured values for the test:

Table 5.2: Measured illuminance valued versus expected values

Distance	Minimum value	Maximum value	Measured value
5	1 334 LUX	5 669 LUX	T.B.D
100	6.66 LUX	28.32 LUX	T.B.D

An Illuminance meter will be utilized to measure the illuminance of the laparoscope's light source.

The final test is to validate the heat transfer model, developed in chapter 4. To perform this test, the built product will be turned on for a long duration of time (8 hours) and the temperature in the laparoscope at points of interest will be measured using the built-in sensors. The measured values will be compared to expected values to determine the accuracy of the heat transfer model.

5.2 Specification Tests

Specification tests are required to determine values and relationships of interest of the laparoscope. For the laparoscope there are two tests that fall into this category: heat dissipation test of the camera module and light source; and digital thermocouples' calibration tests.

The developed product makes use of two digital thermocouples to measure the temperature inside the laparoscope. These thermocouples will be placed in the tip and handle of the product. However, before these components can be placed inside the laparoscope, they need to be calibrated to give the correct measurements. Therefore, the following test was executed.

The digital thermocouples and a calibrated thermocouple were placed within water, that was either heated or frozen. In five minute intervals, the temperature of the water was measured through all the available sensors, until within 5 °C of room temperature. The measured data was then plotted, and the best linear fits were found for each group of data points. Figure 5.1 and Figure 5.2 below shows the digital sensors measurements against the calibrated sensor measurements:

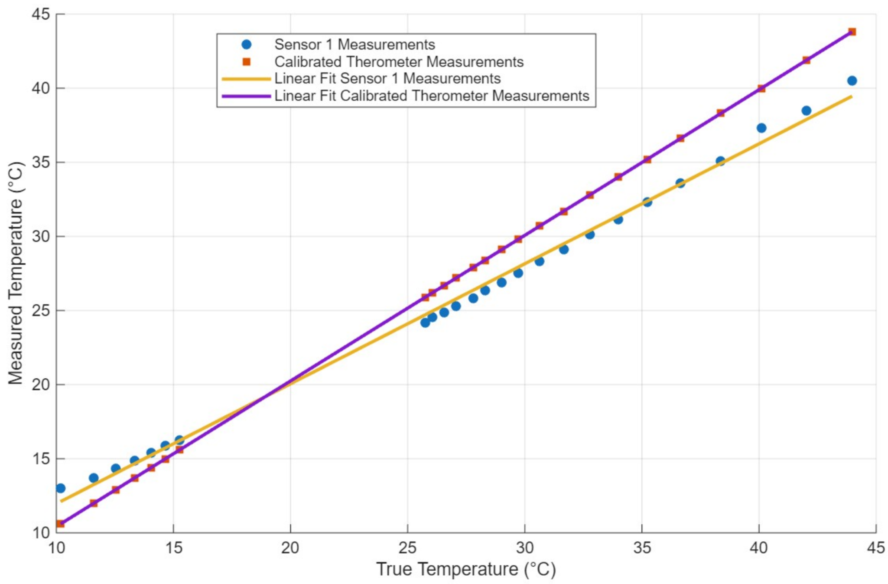


Figure 5.1: Sensor one vs calibrated sensor

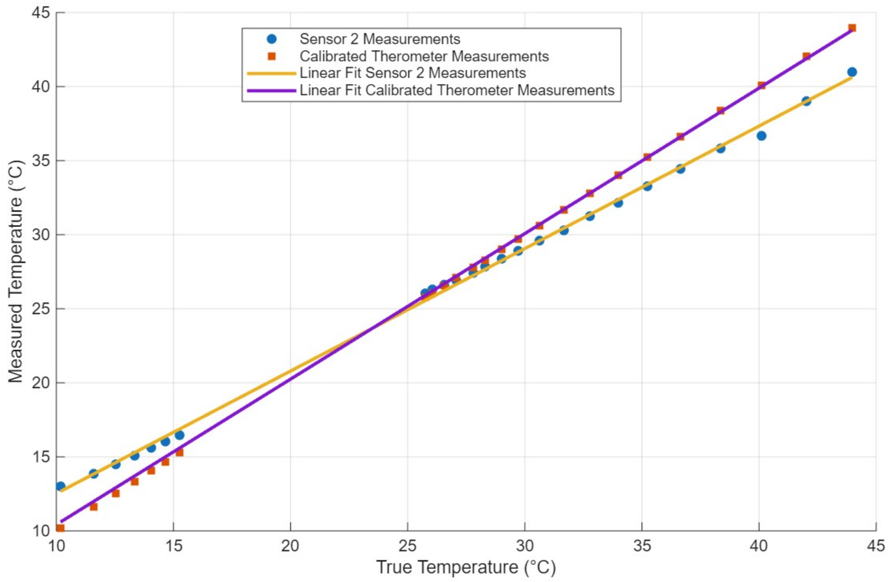


Figure 5.2: Sensor two vs calibrated sensor

Based on the information extracted from the graphs shown above, it appears that the sensor readings have a linear relationship with the true temperature measurements. Therefore, the measured temperatures can be easily converted into the true temperature with the help of the following equations:

$$Measurement_1 = (Sensor_1(Reading) - 3.8676)/0.8095 \tag{5.2a}$$
$$Measurement_2 = (Sensor_2(Reading) - 4.2511)/0.8271 \tag{5.2b}$$

The equations mentioned above are derived using the linear polyfit function from Mathworks Matlab.

The second specifications test required for the laparoscope, is to determine how much heat is dissipated by the laparoscope's camera module and light source. This value is important, because it will determine whether the design of the laparoscope will be able to dissipate the generated heat effectively or not. It is also critical to for the heat transfer model.

To determine how much heat is generated, the camera module and the light source are placed inside a thermal container, with a temperature sensor. The container is then sealed, while the module and light source are turned on. Over two hours, the temperature is periodically measured every ten minutes. The captured data is then processed, and the following information was extracted:

- Average heat dissipated = 0.32606 μW
- Maximum heat dissipated = 3.8865 μW
- Minimum heat dissipated = - 2.0298 μW

However, looking at the information extracted from the test, it shows unexpected data. Especially, the minimum heat dissipated value, which is negative. This means for a period of time, the container got colder instead of hotter. This could be due to inaccuracies of the sensors, because they are only accurate with 0.5 °C tolerance. However, looking at the Figure 5.3 below, it shows that it is not just a sensor issue. The most likely explanation for this result is that the container was not properly sealed, which lead to heat escaping the container, as well as let cold air enter the container. Therefore, the experiment will need to be performed again.

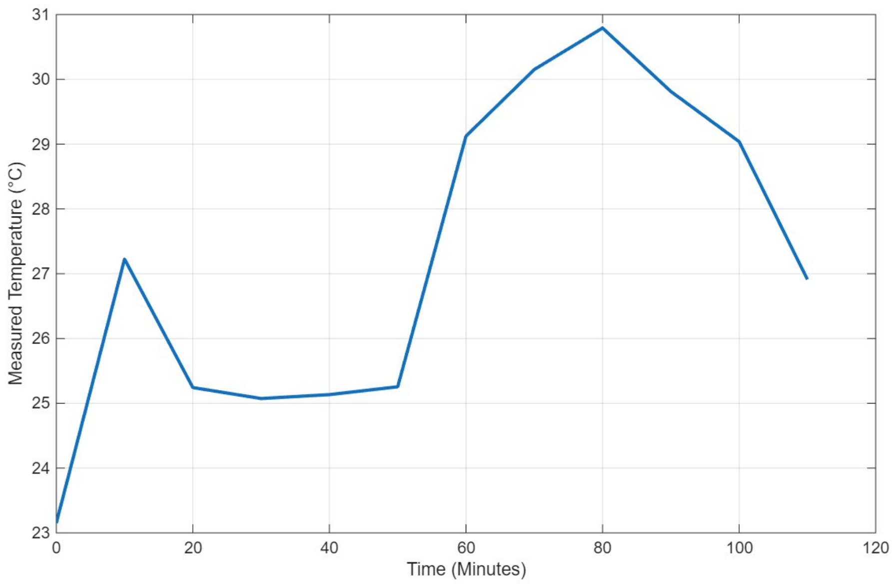


Figure 5.3: Measured temperature vs time graph

5.2.1 Functionality Test

The functionality tests are the test required to be performed to determine whether the developed product functions as a medical grade laparoscope. These are tests that require a testing station, or a physical box trainer. These test will be performed when the product is built and will illustrate if the product functions as desired.

Chapter 6

Conclusions

6.1 Conclusion

Laparoscopy is a rapidly expanding industry in the medical field. It is an innovative approach to minimize the harm surgeons inflict on the patients. However, it is a more complex procedure to perform, thus requiring more training. Consequently, there is a desire for medical robots to assist surgeons during laparoscopic surgery.

Therefore, there is a need for a low-cost laparoscope for training purposes. Consequently, it can be used for prototype testing of new medical robots. This low-cost laparoscope is to be tested to determine its functionality. Therefore, a testing station is required to be developed to determine the laparoscope's capabilities.

For this reason, this project aimed to develop a low-cost laparoscope in addition to a testing station.

List of references

ArduCam (2020). *Arducam 12MP×2 synchronized stereo camera bundle kit for Raspberry Pi and Pi Zero, two 12.3MP IMX477 camera modules with CS lens and ArduCam CamArray stereo camera HAT*. [Online]. Accessed: [25/05/2026].
Available at: https://www.arducam.com/arducam-12mp2-synchronized-stereo-camera-bundle-kit-for-raspberry-pi-two-12-3mp-imx477-camera-modules-with-cs-lens-and-arducam-camarray-stereo-camera-hat.html

ArduCAM (2022). ArduCAM/Arduino. [Online]. Accessed: [25/05/2026].
Available at: https://github.com/ArduCAM/Arduino

Arduino AG (2026*a*). Language reference. [Online]. Accessed: [25/05/2026].
Available at: https://docs.arduino.cc/language-reference/#structure

Arduino AG (2026*b*). Products. [Online]. Accessed: [25/05/2026].
Available at: https://store.arduino.cc/products

Buia, A., Stockhausen, F. and Hanisch, E. (2015). Laparoscopic surgery: A qualified systematic review. *World Journal of Methodology*, vol. 5, no. 4, pp. 238–254.

Cambridge in Colour (2020*a*). Camera exposure. [Online]. Accessed: [25/05/2026].
Available at: https://www.cambridgeincolour.com/tutorials/camera-exposure.htm

Cambridge in Colour (2020*b*). Digital camera sensors. [Online]. Accessed: [25/05/2026].
Available at: https://www.cambridgeincolour.com/tutorials/camera-sensors.htm

EBME and Clinical Engineering Articles (2026). Endoscope design and assembly. [Online]. Accessed: [25/05/2026].
Available at: https://www.ebme.co.uk/articles/clinical-engineering/endoscope-design-and-assembly

Ibrahim, A., Ulrich, M., Liselotte, M., Julian, P., Nicolai, M., Matthias, B., Georgios, G., Antonio Simone, L. and Damaris, F. (2021). The development of laparoscopy — a historical overview. *Frontiers in Surgery*, vol. 8.

International Electrotechnical Commission (2021). Medical electrical equipment — part 2-41: Particular requirements for the basic safety and essential performance of

surgical luminaires and luminaires for diagnosis. Tech. Rep. IEC 60601-2-41, International Electrotechnical Commission.
Available at: https://webstore.iec.ch/en/publication/62677

International Organization for Standardization (2025). Endoscopes — medical endoscopes and endotherapy devices. Tech. Rep. ISO 8600-1:2025, International Organization for Standardization.

Intuitive Surgical (2026). Da Vinci SP. [Online]. Accessed: [25/05/2026].
Available at: https://www.intuitive.com/en-us/products-and-services/da-vinci/sp

Karl Storz Endoskope (2026*a*). Home page. [Online]. Accessed: [25/05/2026].
Available at: https://www.karlstorz.com/de/en/index.htm

Karl Storz Endoskope (2026*b*). TIPCAM1 RUBINA. [Online]. Accessed: [25/05/2026].
Available at: https://www.karlstorz.com/za/en/product-detail-page.htm?productID=1000108799

Leopard Imaging (2026). Hummingbird mobile camera modules. [Online]. Accessed: [25/05/2026].
Available at: https://leopardimaging.com/hummingbird/

Micro Robotics (Pty) Ltd (2026). Home page. [Online]. Accessed: [25/05/2026].
Available at: https://www.robotics.org.za/

Morrow, D. (2026). Ultimate depth of field photography guide 2026. [Online]. Accessed: [25/05/2026].
Available at: https://www.davemorrowphotography.com/depth-of-field-photography

Olympus Corporation (2026). ENDOEYE 3D 10 mm. [Online]. Accessed: [25/05/2026].
Available at: https://www.olympus.com.au/medical/en/Products-and-Solutions/Products/Product/ENDOEYE-3D-10-mm.html

Raspberry Pi Ltd (2026*a*). Documentation. [Online]. Accessed: [25/05/2026].
Available at: https://www.raspberrypi.com/documentation/

Raspberry Pi Ltd (2026*b*). Products. [Online]. Accessed: [25/05/2026].
Available at: https://www.raspberrypi.com/products/

Schrader, S., Oginski, S. and Brueggemann, D. (2016). [endoscope]. (Patent: US9510744B2). [Online]. Available: https://patents.google.com/patent/US9510744B2 [25/05/2026].

Shenzhen VisionMeta Technology Co., Ltd (2026). 1MP 3.3/3.5/3.9mm OV9734 HD endoscope medical camera module. [Online]. Accessed: [25/05/2026].
Available at: https://www.visionmeta.com/product-category/feature/endoscope/endoscope-diameter/3-5mm-endoscope/

Stellenbosch University (2026). Study guide. [Online]. Accessed: [25/05/2026]. Available at: https://stemlearn.sun.ac.za/my/

Subido, E., Pacis, D.M.M. and Bugtai, N.T. (2018). Recent technological advancements in laparoscopic surgical instruments. *AIP Conference Proceedings*, vol. 1933, no. 1.

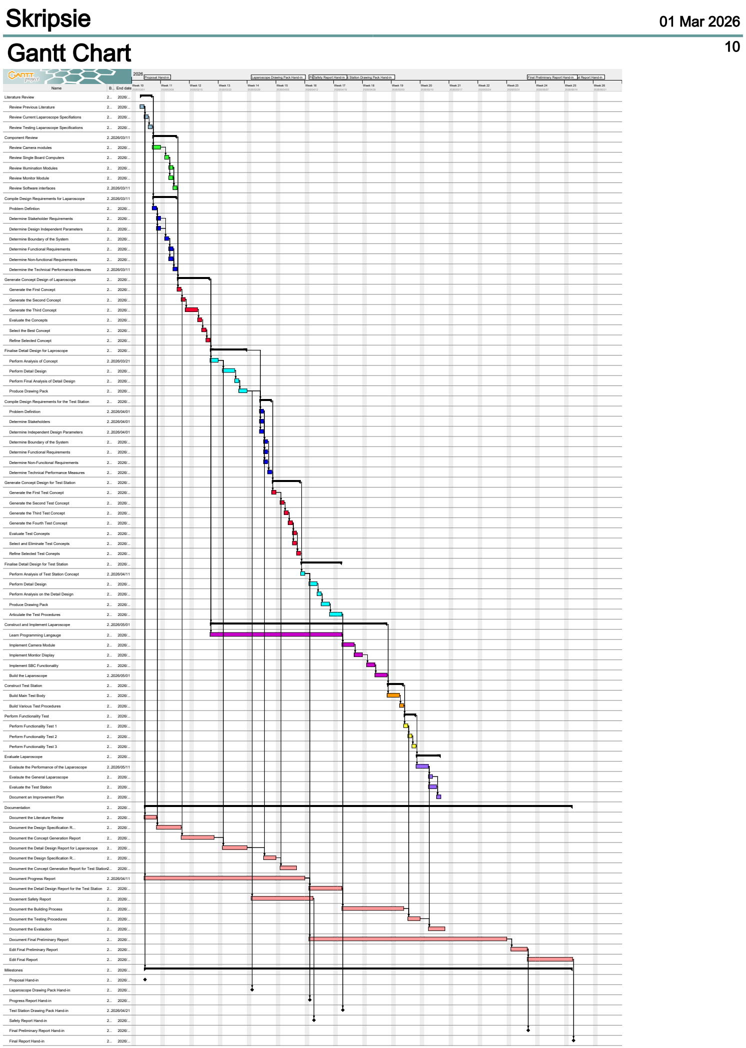
Weather & Climate (2026). Stellenbosch temperature by month. [Online]. Accessed: [25/05/2026].
Available at: https://weather-and-climate.com/average-monthly-min-max-Temperature,stellenbosch,South-Africa

World Standards (2025). South africa — power plug, socket & mains voltage in south africa. [Online]. Accessed: [25/05/2026].
Available at: https://www.worldstandards.eu/electricity/plug-voltage-by-country/south-africa/

Xiao, D., Jakimowicz, J.J., Albayrak, A., Buzink, S.N., Botden, S.M.B.I. and Goossens, R.H.M. (2014). Face, content, and construct validity of a novel portable ergonomic simulator for basic laparoscopic skills. *Journal of Surgical Education*, vol. 71, no. 1, pp. 65–72.

Zumtobel (2018). *The Lighting Handbook*. Zumtobel. [Online]. Available: [URL] [25/05/2026].

Schedule



Estimated Cost per Activity

The estimated cost per activity, shows all the planned activities of the project and how much time would be used to perform each activity. Each activity duration is determined by the gantt chart. It is assumed that there are 8 hours available each working day. And the workload is divided based on importance for each day. Usually 5 hours for planned activities and then 1 hour each day for documentation. This is not a concrete value and is subject to change.

Activity	Engineering Time		Running Cost	Facility Use	Capital Cost	MMW			Total
						Labour		Material	
	hr	R	R	R	R	hr	R	R	R
Literature Review	15	6750	150						6900
Component Review	20	9000	150						9150
Compile Design Requirements for Laparoscope	20	9000	150						9150
Generate Concept Design for Laparoscope	35	15750	150						15900
Finalize Detail Design for Laparoscope	40	18000	150						18150
Compile Design Requirements for Test Station	20	9000	150						9150
Generate Concept Design for Test Station	30	13500	150						13650
Finalize Detail Design for Test Station	40	18000	150						18150
Construct and Implement Laparoscope	95	42750	600	6000	4000	15	4500	500	58350
Construct Test Station	20	9000	600	2000		40	12000	300	23900
Perform Functionality Test	40	18000	600	2000					20600
Evaluate Laparoscope	25	11250	150						11400
Documentation	120	54000	150						54150
Total	**520**	**234000**	**3300**	**10000**	**4000**	**55**	**16500**	**800**	**268600**

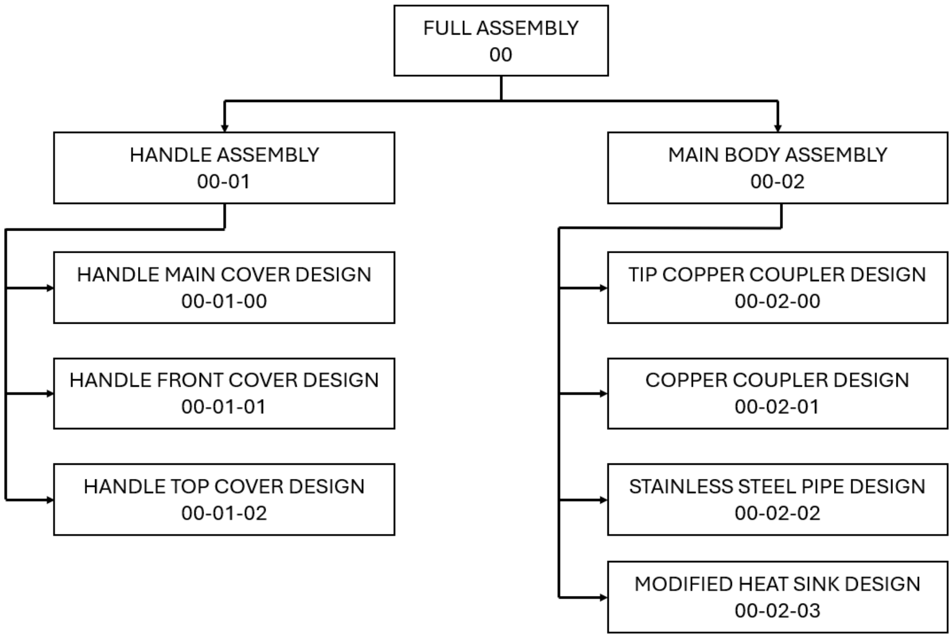
GA Self Assessment

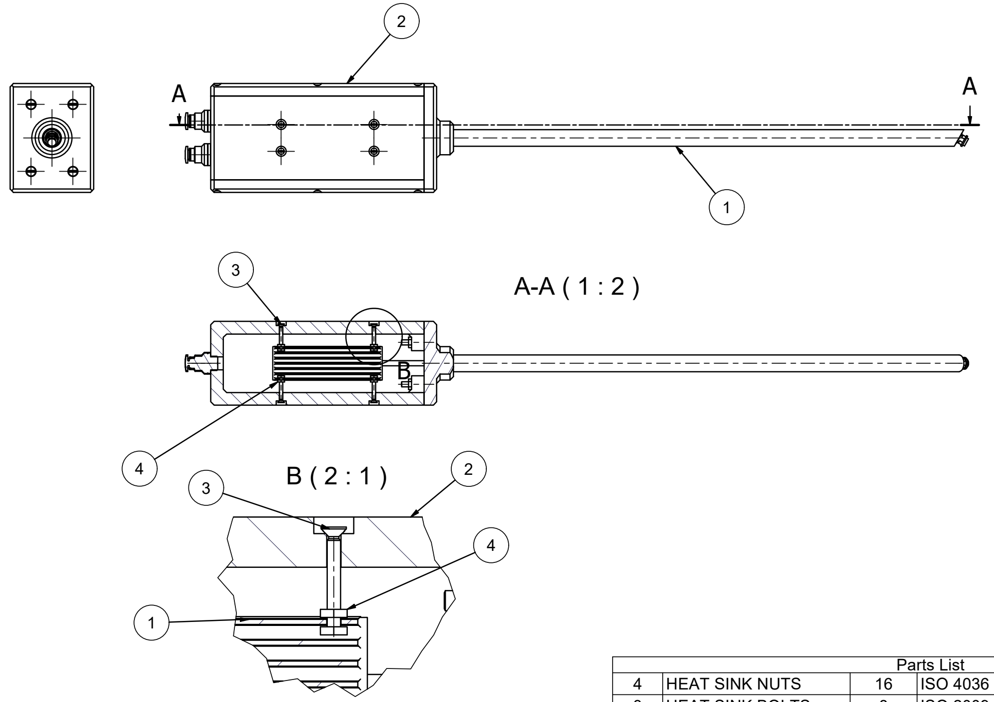
Graduate Attribute	How it will be satisfied
GA 1 - Problem-solving: Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development.	This GA will be satisfied during the literature review, design specification reports, concept generation reports and the detail design reports, including drawing packs.
GA 2 - Application of scientific and engineering knowledge: Apply knowledge of mathematics, natural science, computing and engineering fundamentals, and an engineering specialization to develop solutions to complex engineering problems.	GA 2 will be satisfied during sections where component specification and selection will be conducted. It will also be satisfied during detail design sections where analysis of the design will be performed.
GA 3 - Engineering Design: Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs.	This GA will be satisfied by following the workplan and planned schedule of the project. It will also be satisfied during component selection, detail design process and the evaluation of the developed product.

Continued on next page...

Graduate Attribute (continued)	How it will be satisfied (continued)
GA 5 - Use of Engineering Tools: Demonstrate competence to create, select and apply and recognize limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	GA 5 will be satisfied during the detail design of the project. It will also be satisfied during the implementation of the laparoscope.
GA 6 - Professional and technical communication: Demonstrate competence to communicate effectively and inclusively on complex engineering activities, both orally and in writing, with the engineering community and society at large, taking into account cultural, language, and learning differences.	GA 6 will be satisfied during the documentation of the project, including the progress and final report. It will also be satisfied during the weekly meetings, progress meeting and final presentation.
GA 8 - Individual and collaborative teamwork: Demonstrate competence to function effectively as an individual and as a member or leader in diverse and inclusive teams, and in multi-disciplinary, face-to-face, remote and distributed settings.	This GA will be satisfied during the construction of the laparoscope and testing station. It will also be satisfied during the testing stages of the laparoscope.
GA 9 - Independent Learning Ability: Demonstrate competence to engage in independent learning through well-developed learning skills.	This will be satisfied during the entirety of the project, but predominantly during the literature review.

Drawings


FULL ASSEMBLY
00
HANDLE ASSEMBLY
00-01
MAIN BODY ASSEMBLY
00-02
HANDLE MAIN COVER DESIGN
00-01-00
HANDLE FRONT COVER DESIGN
00-01-01
HANDLE TOP COVER DESIGN
00-01-02
TIP COPPER COUPLER DESIGN
00-02-00
COPPER COUPLER DESIGN
00-02-01
STAINLESS STEEL PIPE DESIGN
00-02-02
MODIFIED HEAT SINK DESIGN
00-02-03



Parts List			
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
4	HEAT SINK NUTS	16	ISO 4036 M2
3	HEAT SINK BOLTS	8	ISO 2009 M2 X 16 X 14
2	HANDLE ASSEMBLY	1	00-01
1	MAIN BODY ASSEMBLY	1	00-02

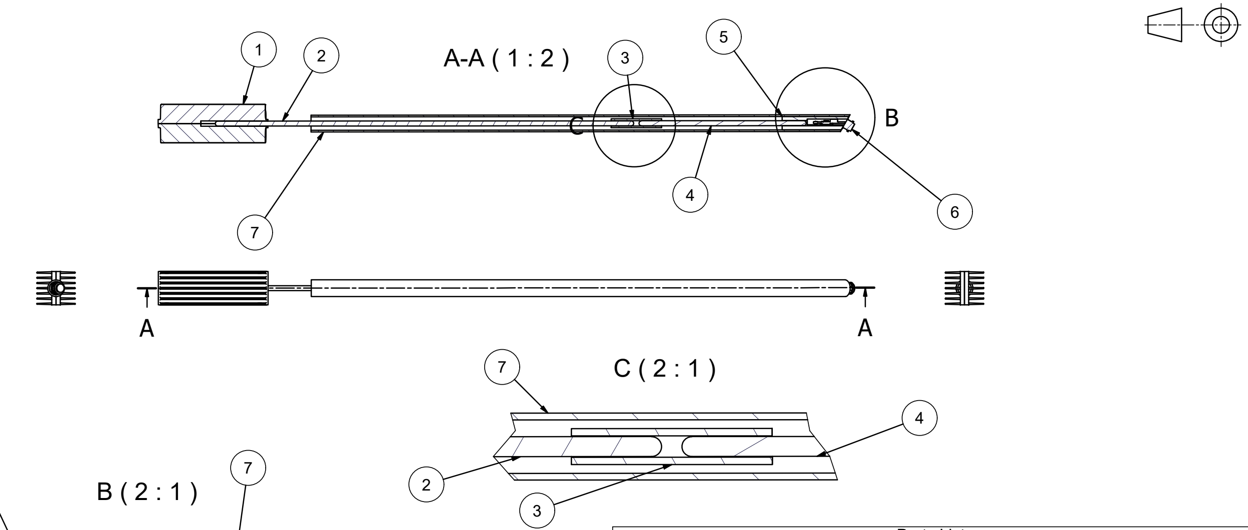
UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1˚

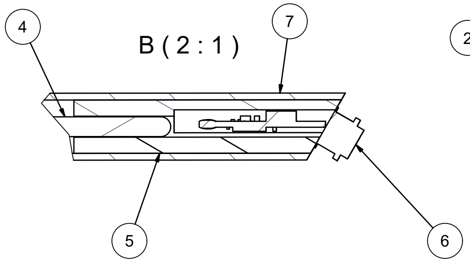
IF IN DOUBT ASK

PW Breytenbach
26122146@sun.ac.za
078 780 9388

STELLENBOSCH UNIVERSITY

SCALE ON A 3 1:4	TITLE: FULL ASSEMBLY		
UNITS IN mm			
STUDENT Nr. 26122146	DRAWN BY PW Breytenbach	CHECKED	
DATE 07/04/2026	SHEET Nr. 1 OF	SHEETS 12	Nr. 00





Parts List			
7	STAINLESS STEEL PIPE	1	00-02-02
6	CAMERA MODULE	1	MICROROBOTICS ENDOSCOPE
5	TIP COPPER COUPLER	1	00-02-00
4	HEAT PIPE - 100 MM	1	SPREADFAST Ø3 MM - 100 MM HEAT PIPE
3	COPPER COUPLER	1	00-02-01
2	HEAT PIPE - 250 MM	1	SPREADFAST Ø3 MM - 250 MM HEAT PIPE
1	MODIFIED HEAT SINKS	2	00-02-03
ITEM	**DESCRIPTION**	**Qty.**	**MATERIAL / SPECIFICATIONS**

UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1˚

IF IN DOUBT ASK
PW Breytenbach
26122146@sun.ac.za
078 780 9388

STELLENBOSCH UNIVERSITY

SCALE ON A 3 1:2	**TITLE:** MAIN BODY ASSEMBLY		
UNITS IN mm			
STUDENT Nr. 26122146	**DRAWN BY** PW Breytenbach	**CHECKED**	**DATE** 07/04/2026
SHEET Nr. 7 **OF** 12 **SHEETS**	**Nr.** 00-02		